

Problem-Solution Fit

Date	29 June 2026
Team ID	SWTID-2026-4993
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

The Problem–Solution Fit Canvas is used to analyze customer needs, identify key problems, and develop an effective solution. It ensures the project addresses real-world challenges and delivers value to users..

1. CUSTOMER SEGMENT(S) [CS] • Small-scale & marginal farmers • Agricultural researchers • Agribusiness organizations • Government policymakers	6. CUSTOMER LIMITATIONS [CL] • Limited access to agricultural experts • Low digital literacy in rural areas • Budget constraints for tools • Poor Internet in remote regions	5. AVAILABLE SOLUTIONS [AS] • Pros: Govt. advisory services exist • Pros: General crop calendars available • Cons: Not personalized to soil data • Cons: Not ML-powered or real-time
2. PROBLEMS / PAINS [PR] • Wrong crop selection → losses (Frequent) • Inefficient use of N, P, K (Daily) • No soil-climate matching tool (Always) • Reduced yield due to poor planning	9. PROBLEM ROOT / CAUSE [RC] • No centralized ML crop system • Farmers lack soil chemistry knowledge • Existing tools are generic only • Agricultural data is unprocessed	7. BEHAVIOR + INTENSITY [BE] • Consults neighbours for advice (High) • Relies on experience not data (High) • Avoids tech due to complexity (Med) • Seeks simple recommendations (High)
3. TRIGGERS TO ACT [TR] • Crop failure in previous season • Rising fertilizer & water costs • Government push for smart farming • Peer success with tech adoption	10. YOUR SOLUTION [SL] • OptiCrop AI/ML Recommendation System • Input: N, P, K, Temp, Humidity, pH, Rainfall • ML: Logistic Regression & K-Means • Output: Best crop + confidence score • Flask Web App: Simple accessible UI	8. CHANNELS OF BEHAVIOR [CH] ONLINE: • Flask Web Application • GitHub Repository • Digital agri portals OFFLINE: • Field demonstrations • Farmer training workshops
4. EMOTIONS BEFORE / AFTER [EM] • BEFORE: Anxious about crop failure • BEFORE: Uncertain about soil suitability • AFTER: Confident with data-driven advice • AFTER: Motivated to adopt smart farming		

